

surface, its acceleration will be less than 9.81 m/s^2 . Option D.

12	A weather balloon, initially stationary on the ground, contains an ideal gas with an internal energy of 500 J. The balloon is then launched into the air, reaching a certain altitude Y where the potential energy of the gas has increased by 100 J and its kinetic energy is 250 J. Assuming the balloon is tied tightly and there is no heat transfer or work done on the ideal gas, what is the internal energy of the gas when the balloon is at that altitude Y?			
	A 150 J	B 350 J	C 500 J	D 850 J
	Answer: C Given $Q = 0$ and $W = 0$, thus $\Delta U = 0$. No change in internal energy $\rightarrow$ internal energy remains at the initial value of 500 J. Recall also that internal energy of an ideal gas is associated with the kinetic energy due to the random motion of the molecules, and independent of the bulk changes in KE and PE of the system as a whole.			

13	A sealed container is filled with an ideal gas which is initially at a temperature of 50°C . The
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